

Assignment - 2

**Course: MST501-1: Python
Programming for Statistics**



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1. Design a Python program for a monthly budget planner. The user enters monthly income, fixed expenses, variable expenses, and savings percentage. Use arithmetic, comparison, and logical operators to calculate total expenses, expected savings, remaining balance, and determine whether the user is staying within the recommended savings goal.

Ans-

```
mi=float(input("Enter monthly income"))
fe=float(input("Enter fixed expenses"))
ve=float(input("Enter variable expenses"))
sp=float(input("Enter desired savings percentage"))
te=fe+ve
es=(mi*sp)/100
rb= mi-te
print("Monthly income is",mi)
print("Total Expenses is",te)
print("Expected Savings is",es)
print("Remaining Balance is",rb)
if rb>es:
    print("You are meeting your savings goal more than you expect")
elif rb==es:
    print("You are exactly meeting your savings goal")
else:
    print("You are spending more")
```

```
Py2.py - C:\Users\Rajku\OneDrive\Desktop\Python\Py2.py (3.14.5)
File Edit Format Run Options Window Help
mi=float(input("Enter monthly income"))
fe=float(input("Enter fixed expenses"))
ve=float(input("Enter variable expenses"))
sp=float(input("Enter desired savings percentage"))
te=fe+ve
es=(mi*sp)/100
rb= mi-te
print("Monthly income is",mi)
print("Total Expenses is",te)
print("Expected Savings is",es)
print("Remaining Balance is",rb)
if rb>es:
    print("You are meeting your savings goal more than you expect")
elif rb==es:
    print("You are exactly meeting your savings goal")
else:
    print("You are spending more")

IDLE Shell 3.14.5
File Edit Shell Debug Options Window Help
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1916 AMD64] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Rajku\OneDrive\Desktop\Python\Py2.py =
Enter monthly income65000
Enter fixed expenses25000
Enter variable expenses15000
Enter desired savings percentage15
Monthly income is 65000.0
Total Expenses is 40000.0
Expected Savings is 9750.0
Remaining Balance is 25000.0
You are meeting your savings goal more than you expect
>>>
===== RESTART: C:\Users\Rajku\OneDrive\Desktop\Python\Py2.py =
Enter monthly income65000
Enter fixed expenses25000
Enter variable expenses15000
Enter desired savings percentage15
Monthly income is 65000.0
Total Expenses is 40000.0
Expected Savings is 9750.0
Remaining Balance is 25000.0
You are meeting your savings goal more than you expect
>>>|
```

2. Develop a Python program that calculates a household electricity bill using slab-based pricing. The program should apply different rates based on units consumed, add fixed charges and taxes, and determine whether the household qualifies for an energy-efficiency rebate. Use appropriate data types, nested expressions, and multiple operators to generate a detailed bill summary.

Ans-

```
name=input("Enter your name")
units=int(input("Enter units consumed"))
fixedcharge=75
taxrate=3
if units <= 25:
    bill = units * 1.75
    print("Your bill is",bill)
elif units <= 50:
    bill = (25 * 1.75) + ((units - 25) * 2.75)
```

```

    print("Your bill is",bill)
else:
    bill=(25*1.75) + (25*2.75) + ((units -50)*3.75)
    print("Your bill is",bill)
tax =(bill*taxrate)/100
total= bill + fixedcharge + tax
print("Tax paid",tax)
print("Total bill",total)
if units <150:
    rebate =total*0.10
    final =total- rebate
    print("Eligible for 10.25% rebate")
else:
    rebate= 0
    final= total
    print("Not eligible for rebate")

```

The screenshot shows a Python IDE with two windows. The left window displays the source code for a program that calculates a bill based on units consumed. The right window shows the output of the program, including user input and calculated values.

Py2.py - C:\Users\Rajku\OneDrive\Desktop\Python\Py2.py (3.14.5)

```

File Edit Format Run Options Window Help
name=input("Enter your name")
units=int(input("Enter units consumed"))
fixedcharge=75
taxrate=3
if units <= 25:
    bill = units * 1.75
    print("Your bill is",bill)
elif units <= 50:
    bill = (25 * 1.75) + ((units - 25) * 2.75)
    print("Your bill is",bill)
else:
    bill =(25*1.75) + (25*2.75) + ((units -50)*3.75)
    print("Your bill is",bill)
tax =(bill*taxrate)/100
total= bill + fixedcharge + tax
print("Tax paid",tax)
print("Total bill",total)
if units <150:
    rebate =total*0.10
    final =total- rebate
    print("Eligible for 10.25% rebate")
else:
    rebate= 0
    final= total
    print("Not eligible for rebate")

```

IDLE Shell 3.14.5

```

File Edit Shell Debug Options Window Help
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Rajku\OneDrive\Desktop\Python\Py2.py =====
Enter your nameAbkita Das
Enter units consumed250
Your bill is 862.5
Tax paid 25.875
Total bill 963.375
Not eligible for rebate
>>>

```

3. Create a Python program for an online shopping checkout system. The user enters product costs, quantities, membership status, coupon eligibility, and payment method. The program must calculate subtotal, tiered discounts, loyalty rewards, taxes, shipping charges, and final payable amount using complex expressions and operator combinations. It should also determine eligibility for premium membership upgrades based on spending patterns and reward points earned.

Ans-

```
name =input("Enter your name")
cost =int(input("Enter product cost"))
quantity =int(input("Enter quantity"))
member =input("Are you a member")
coupon =input("Do you have a coupon")
payment =input("Your payment method")
subtotal =cost*quantity
print("Subtotal", subtotal)
if subtotal >=5570:
    discount = subtotal * 0.20
    print("Discount:", discount)
elif subtotal >= 2750:
    discount = subtotal * 0.10
    print("Discount:", discount)
else:
    print("No Discount")
if coupon == "yes":
    discount += subtotal * 0.05
if member == "yes":
    reward= subtotal // 100
    print("Reward Points", reward)
else:
    reward =0
```

```

    print("Reward Points", reward)

tax =(subtotal -discount)*0.08

print("Tax", tax)

if subtotal >=3750:

    shipping =150

else:

    shipping =250

final =subtotal- discount+tax +shipping

print("Final Amount",final)

if (subtotal >= 5570) and (reward>= 50):

    print("Eligible for Premium Membership")

else:

    print("Not Eligible for Premium Membership")

```

The screenshot shows a Python IDE with two windows. The left window, titled 'Py2.py', contains a script that calculates the final amount of a purchase based on subtotal, discount, tax, and shipping. It also checks for premium membership eligibility based on subtotal and reward points. The right window, titled 'IDLE Shell 3.14.5', shows the execution of the script. The user has entered 'Ankita Das' as their name, '3650' as the product cost, '1270' as the quantity, 'yes' as a member, and 'Cash' as the payment method. The output shows a subtotal of 4635.00, a discount of 927.10, a reward of 46.35, a tax of 278.13, and a final amount of 3754.905. The user is eligible for premium membership.

```

Py2.py - C:\Users\Rajku\OneDrive\Desktop\Python\Py2.py (3.14.5)
File Edit Format Run Options Window Help
name=input("Enter your name")
cost =int(input("Enter product cost"))
quantity =int(input("Enter quantity"))
member =input("Are you a member")
coupon =input("Do you have a coupon")
payment =input("Your payment method")
subtotal =cost*quantity
print("Subtotal", subtotal)
if subtotal >=5570:
    discount = subtotal * 0.20
    print("Discount:", discount)
elif subtotal >= 2750:
    discount = subtotal * 0.10
    print("Discount:", discount)
else:
    print("No Discount")
if coupon == "yes":
    discount += subtotal * 0.05
if member == "yes":
    reward= subtotal // 100
    print("Reward Points", reward)
else:
    reward =0
    print("Reward Points", reward)
tax =(subtotal -discount)*0.08
print("Tax", tax)
if subtotal >=3750:
    shipping =150
else:
    shipping =250
final =subtotal- discount+tax +shipping
print("Final Amount",final)
if (subtotal >= 5570) and (reward>= 50):
    print("Eligible for Premium Membership")
else:
    print("Not Eligible for Premium Membership")

IDLE Shell 3.14.5
File Edit Shell Debug Options Window Help
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Rajku\OneDrive\Desktop\Python\Py2.py =====
Enter your nameAnkita Das
Enter product cost3650
Enter quantity1270
Are you a memberyes
Do you have a couponyes
Your payment methodCash
Subtotal 4635.00
Discount: 927.100.0
Reward Points 46.355
Tax 278.130.0
Final Amount 3754.905.0
Eligible for Premium Membership
>>>

```

